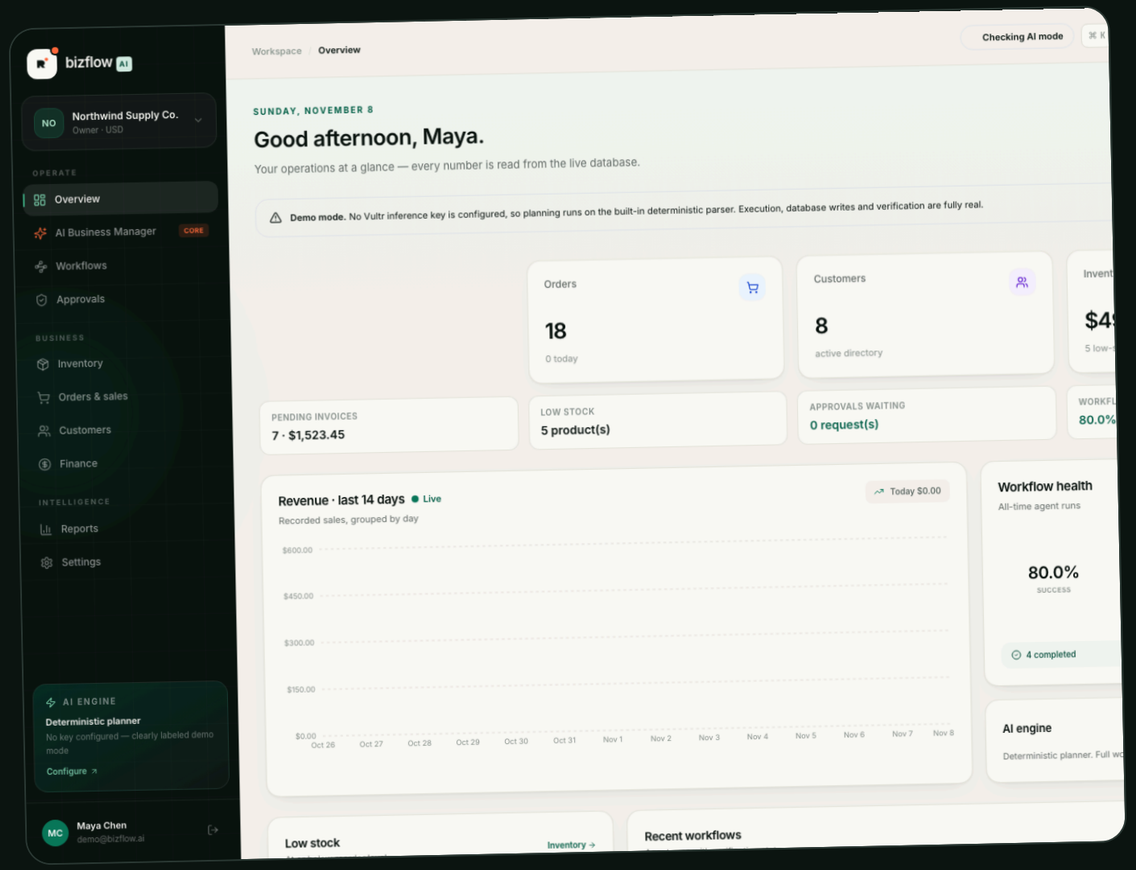
bizflow AI

VULTR AGENT RUSH 2026 · CHALLENGE 02

BizFlow AI

Verified business operations with human control.

Powering the Future of Work



UMAR AKHTAR · LABLAB.AI SUBMISSION

01 · PROBLEM

An order touches four systems. One failure breaks the books.

STEP 01
Customer record

STEP 02
Inventory

FAILS HERE
Sales ledger

NEVER REACHED
Finance

A half-finished write leaves the order created, the stock unchanged and the invoice missing. Nobody notices until month-end.

02 · THE GAP

Most AI tools describe the work. They don't do it.

CHATBOT

- Writes you a plan
- Explains what it would do
- Leaves you to do the work
- Can't prove anything happened

BIZFLOW AI

- **Executes real transactions**
- **Writes atomically — all or nothing**
- **Verifies by reading records back**
- **Stops for a human before anything leaves**

03 · SOLUTION

One sentence in. Real operations out.

01

Plan

Plain-English intent resolved against real customers and products.

02

Execute

Only registered tools run. One atomic transaction writes every record.

03

Verify

Every write is re-read from the database and checked independently.

04

Approve

Messages, refunds and restocks wait for a named human decision.

The model never touches the database directly. It plans; the tools write.

04 · ARCHITECTURE

A tool-bound orchestrator, not a free-roaming agent.



Only registered tools can write. Every call is logged, timed and audited.

05 · WHAT MAKES IT DIFFERENT

It proves the work happened.

Atomic writes

Order, stock, movement, sale and invoice commit together — or not at all.

Verify by read-back

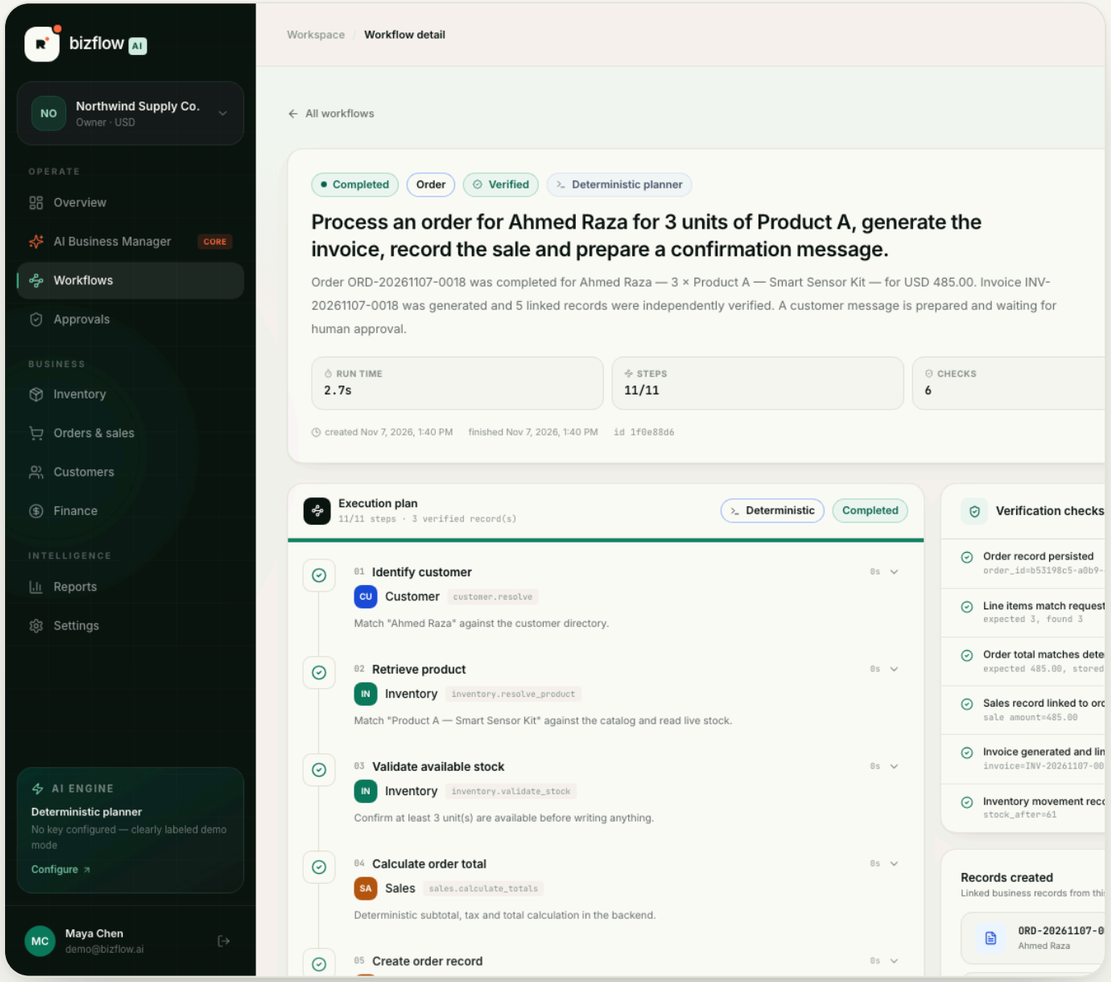
Success is checked against persisted rows, never asserted.

Prepare ≠ execute

Sensitive actions create an approval request, not a side effect.

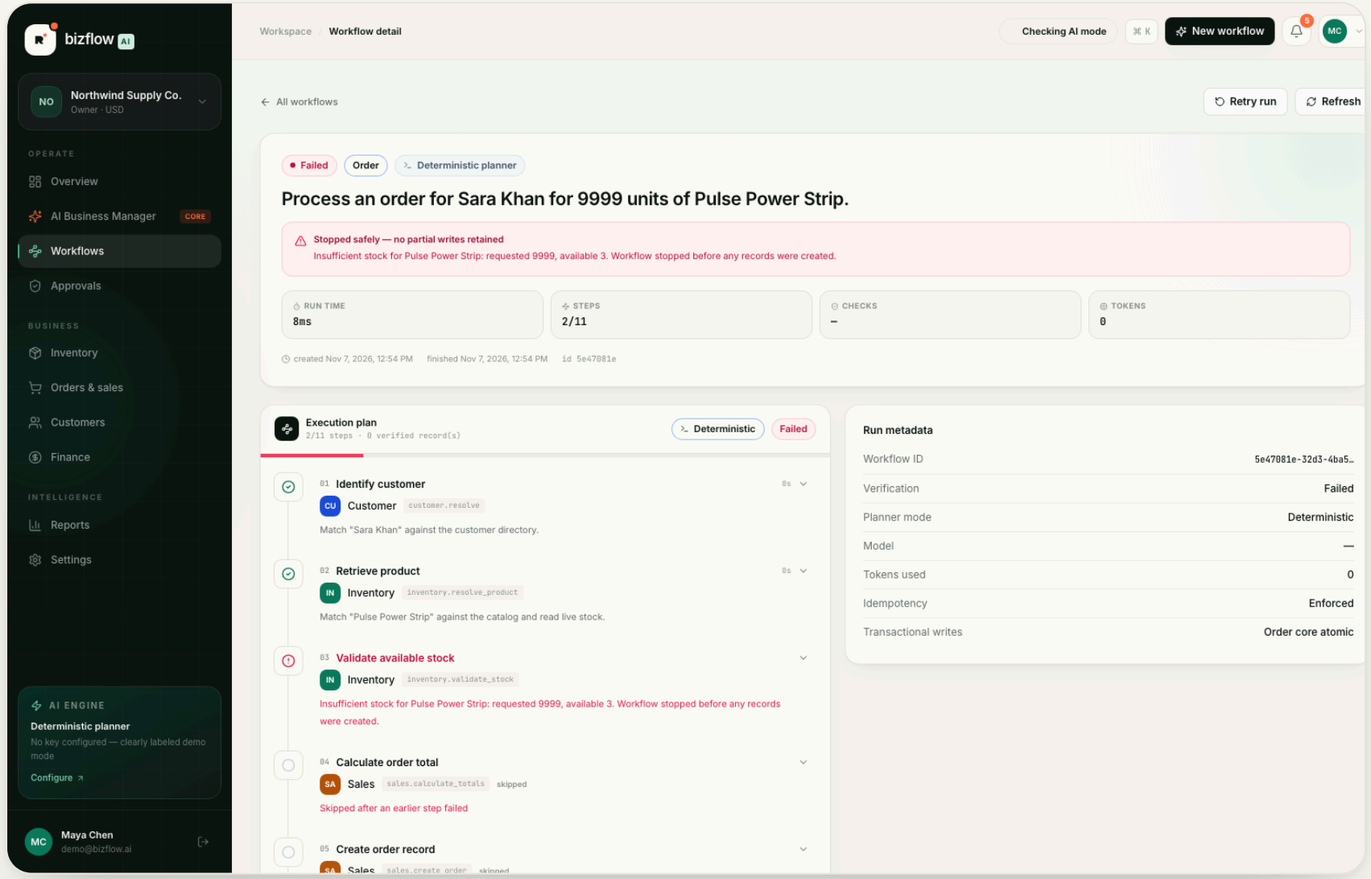
Honest AI mode

The UI states when no model is connected. No fake agent output.



06 · SAFETY

Ask for the impossible. Nothing is written.



STATUS

failed

RECORDS WRITTEN

0

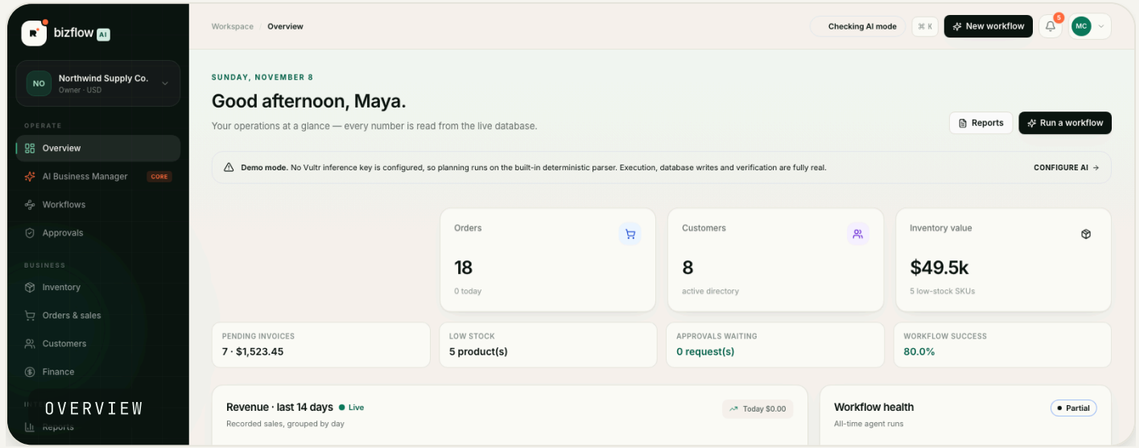
LATER STEPS

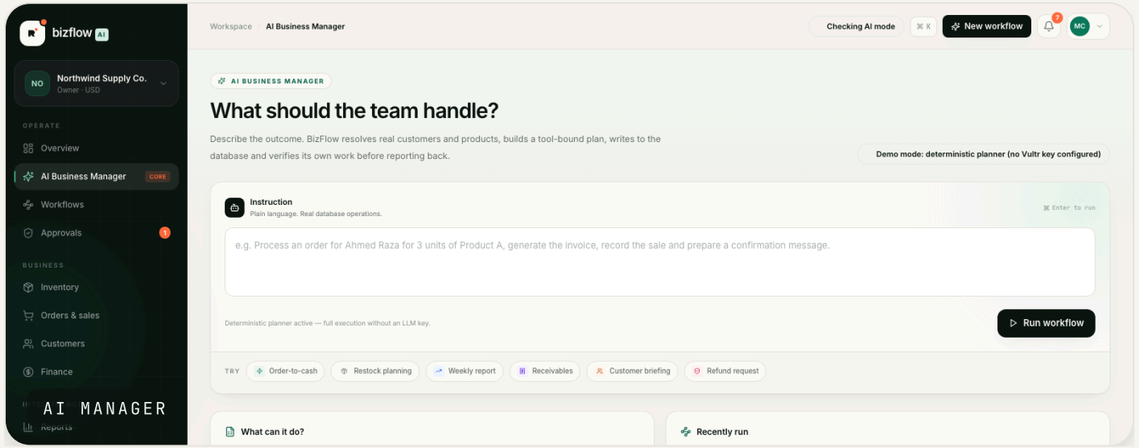
skipped

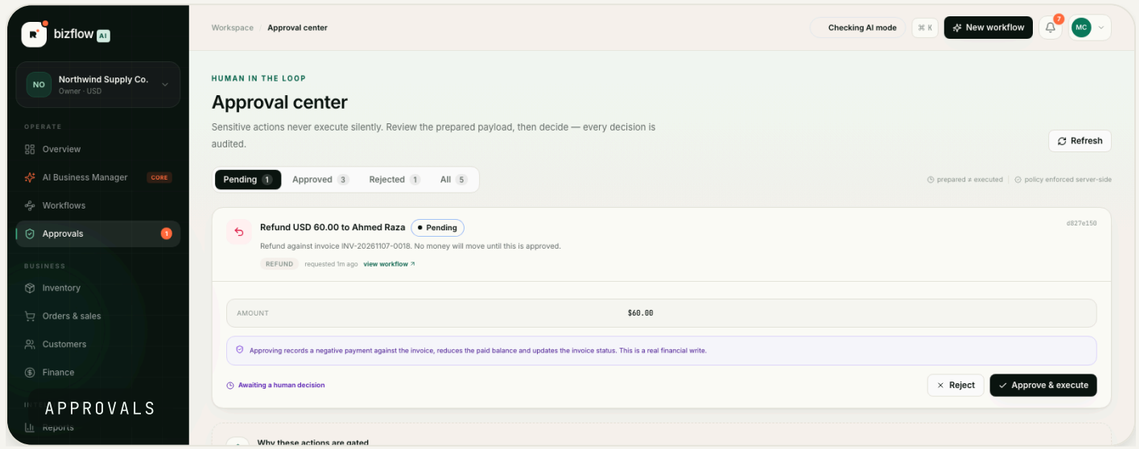
> A system that lies about success is worse than one that fails.

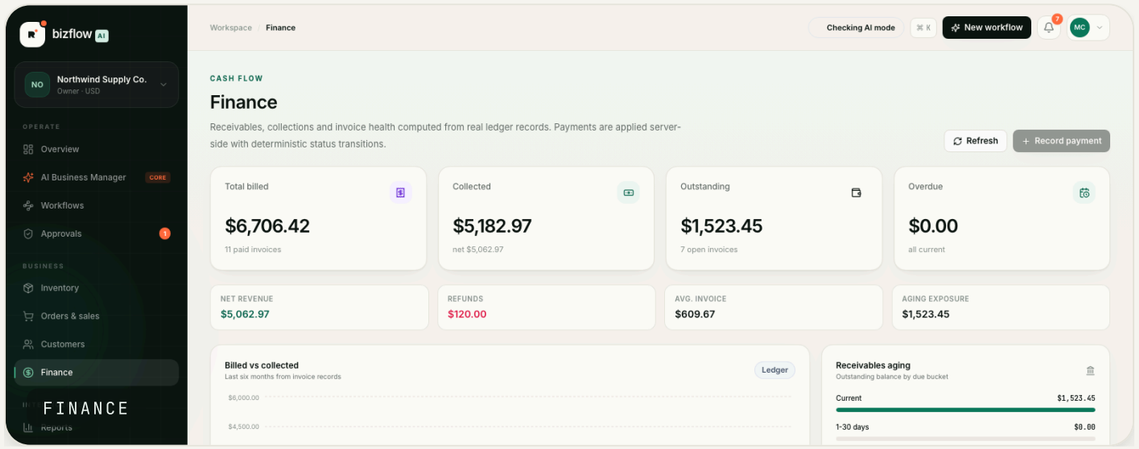
07 · PRODUCT

Fifteen screens.
Every number from the database.









08 · INFRASTRUCTURE

Vultr Cloud Compute. Docker Compose. PostgreSQL 16.

COMPUTE	Vultr Cloud Compute · Ubuntu 24.04
EDGE	Nginx · TLS via Certbot
APPLICATION	FastAPI · Uvicorn · Alembic migrations
DATA	PostgreSQL 16 · persistent volume

```
health       /api/health
migrations   alembic upgrade head
isolation    tenant-scoped queries
audit        append-only log
```

09 · CLOSING

An operations layer you can actually trust.

REAL WRITES

VERIFIED RESULTS

HUMAN CONTROL

repo github.com/umar-akhtar-designer/Bizflow-AI

demo [ADD PUBLIC URL WHEN LIVE]

built for Vultr Agent Rush 2026 · Challenge 02